

Summary: How the Web Works (Step 1)

1. What Happens When You Open a Website

- When you enter a website address (URL), your browser sends a request to a remote server where that website's code (HTML, CSS, JavaScript) is stored. - This frontend code is sent to your browser and runs on your computer, letting you view and interact with the site. - Only the frontend code (not backend or database code) reaches the user's device.

2. What is Frontend, Backend, and Database?

- The **frontend** is the part of the website that users see and interact with (buttons, text, images). It's made with HTML, CSS, and JavaScript. - The **backend** is the server-side part of a website that handles business logic and user requests. Backend code does **not** travel to the user's computer — only frontend code does. - JavaScript on the frontend allows the user to interact with the page (uploading images, typing messages, etc.). These actions are then sent to the backend, which verifies and processes them. - After verification, the backend sends the data (images, messages, etc.) to the **database**, which stores them. - So the backend acts as a **bridge between frontend and database**.

3. Where Do Backend and Database Live?

- Backend, databases, and even the original frontend code live in **server rooms** owned by companies like Facebook. - These rooms have thousands of computers that store and process user data. - So the backend code, database, and frontend all live on these **remote servers** — not on your local computer.

4. How Friends See What You Post

- When you upload an image, it goes from **frontend** → **backend** → **database**. - Each user has an **isolated space** on Facebook's servers containing their info and content. - When you friend someone, Facebook connects your space to theirs. - This connection allows your friend's frontend code to access shared content like your image. - So when your friend loads Facebook, their **frontend code is downloaded** and retrieves your image from the **database** via backend.

5. Security and What Can Be Shared

- Even though there is a connection between your space and your friend's space, not everything is shared. - Sensitive data like **passwords** never get transferred. - The backend contains logic (coded by developers at Facebook) that decides what should be accessible and what should be kept private.